Firewall Management System Requirements

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

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1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks. The system should allow authorized network administrators to create, review, update, and revoke firewall rules, enforce access restrictions between network segments, monitor blocked and allowed traffic, and generate alerts for suspicious or unauthorized communication attempts. The platform should maintain audit logs of firewall policy changes and administrative actions, provide a dashboard for reviewing network security events, and support multiple campus buildings. The solution should be secure, scalable, reliable, and manageable.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Policy	A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Firewall	A control that permits or restricts network traffic according to rules; mentioning it does not authorize automatic changes.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures and the SDLC plan are available in the supporting report.

2 Overall description

2.1 Product perspective

Design a centralized firewall management system for a university that manages firewall policies across academic, administrative, research, hostel, and guest networks.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system - FR-001
- allow authorized network administrators to create firewall rules - FR-002
- allow authorized network administrators to review firewall rules - FR-003
- allow authorized network administrators to update firewall rules - FR-004
- allow authorized network administrators to revoke firewall rules - FR-005
- enforce access restrictions between network segments - FR-006
- monitor blocked and allowed traffic - FR-007
- generate alerts for suspicious or unauthorized communication attempts - FR-008
- maintain audit logs of firewall policy changes and administrative actions - FR-009
- provide a dashboard for reviewing network security events - FR-010
- support multiple campus buildings - FR-011

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
Admin	network administrators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

- The solution should be secure, scalable, reliable, and manageable.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- Which fields must the audit logs contain?
- What retention period is required for the audit logs?
- Which security acceptance outcomes must be demonstrated?
- What measurable scalability targets must the system meet?

- What measurable reliability and recovery targets must the system meet?
- What criteria define acceptable manageability?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

Related requirements: FR-010. Interface details beyond those requirements remain unspecified.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.4 Communications interfaces

Related requirements: FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-010. Interface details beyond those requirements remain unspecified.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall manage firewall policies across academic, administrative, research, hostel, and guest networks in a centralized system.

Traceability: E-001, TEST-001

FR-002 · Should · explicit

The system shall allow authorized network administrators to create firewall rules.

Traceability: E-002, TEST-002

FR-003 · Should · explicit

The system shall allow authorized network administrators to review firewall rules.

Traceability: E-002, TEST-003

FR-004 · Should · explicit

The system shall allow authorized network administrators to update firewall rules.

Traceability: E-002, TEST-004

FR-005 · Should · explicit

The system shall allow authorized network administrators to revoke firewall rules.

Traceability: E-002, TEST-005

FR-006 · Should · explicit

The system shall enforce access restrictions between network segments.

Traceability: E-002, TEST-006

FR-007 · Should · explicit

The system shall monitor blocked and allowed traffic.

Traceability: E-002, TEST-007

FR-008 · Should · explicit

The system shall generate alerts for suspicious or unauthorized communication attempts.

Traceability: E-002, TEST-008

FR-009 · Should · explicit

The system shall maintain audit logs of firewall policy changes and administrative actions.

Traceability: E-003, TEST-009

FR-010 · Should · explicit

The system shall provide a dashboard for reviewing network security events.

Traceability: E-003, TEST-010

FR-011 · Should · explicit

The system shall support multiple campus buildings.

Traceability: E-003, TEST-011

3.3 Performance requirements

NFR-002 · Should · explicit

The system shall be scalable.

Traceability: E-004, TEST-013

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

NFR-003 · Should · explicit

The system shall be reliable.

Traceability: E-004, TEST-014

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-006, FR-009, FR-010.

NFR-001 · Should · explicit

The system shall be secure.

Traceability: E-004, TEST-012

3.6.4 Maintainability

NFR-004 · Should · explicit

The system shall be manageable.

Traceability: E-004, TEST-015

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

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